

Wentao Yu

Ph.D. Candidate in Computer Science

Graph Machine Learning | Federated Learning | Trustworthy AI | Spatio-Temporal Modeling | AI4Science

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Homepage: <https://blgpb.github.io/>

GitHub: <https://github.com/blgpb>

Google Scholar: 310+ citations

RESEARCH INTERESTS

My research focuses on graph machine learning, federated learning, heterogeneous data modeling, trustworthy and robust machine learning, spatio-temporal prediction, multimodal learning, and AI for Science. I am particularly interested in learning from distributed, heterogeneous, privacy-constrained graph and time-series data, with applications in recommendation, risk modeling, urban computing, molecular property prediction, remote sensing, and intelligent perception systems.

EDUCATION

Nanjing University of Science and Technology, School of Computer Science and Engineering

Ph.D. Candidate in Computer Science, expected 2027 | Nanjing, China

Advisor: Prof. Chen Gong

Research: Graph Federated Learning, Personalized Federated Learning, Graph Neural Networks, Heterogeneity Modeling

Xidian University, School of Computer Science

M.S. in Computer Science | Xi'an, China | Sep 2018 - Jun 2021

Research: Deep Learning and Remote Sensing Image Understanding

RESEARCH EXPERIENCE

Hong Kong Baptist University, Visiting Ph.D. Student | Hong Kong | Dec 2024 - Dec 2025

Advisor: Prof. Bo Han

- Conduct research on robust and trustworthy machine learning, graph federated learning, and distributed graph representation learning.
- Work on algorithm design, empirical evaluation, paper writing, and cross-institution research collaboration.

RIKEN AIP, Research Intern | Remote | Oct 2022 - Dec 2023

- Worked on AI4Science research involving graph representation learning, molecular property prediction, and multi-granularity structural modeling.
- Developed and evaluated graph-based models for molecular benchmarks using PyTorch-style research workflows.

SELECTED RESEARCH PROJECTS

FedSSA / FedIIH: Heterogeneity-Aware Graph Federated Learning

ICML 2026 first author; AAAI 2025 first author

- Studied graph federated learning under severe feature-distribution and topology-distribution heterogeneity, where raw graph data cannot be centralized due to privacy or institutional constraints.
- Proposed semantic and structural knowledge-sharing frameworks based on variational inference, cluster-level representative distributions, spectral GNNs, spectral energy, and latent factor disentanglement.
- Evaluated the methods on 11 graph federated learning datasets under overlapping and non-overlapping client

partitions. FedSSA consistently outperformed 11 state-of-the-art baselines, while FedIIH improved average performance on heterophilic graphs by 5.79% over the strongest competitor.

- Released reproducible code and experiment configurations for public research use.

FedTPS / FedHINT: Personalized Federated Traffic Flow Prediction

ICDM 2024 Best Student Paper Runner-Up Award; AAAI 2026

- Addressed privacy-preserving traffic prediction across regions where local models cannot directly access cross-region dependencies.
- Designed global/local pattern sharing with discrete wavelet transforms and a server-side traffic pattern bank; further extended the framework with hidden global components, proxy nodes, and adaptive proxy-local dependency recovery.
- Built a transferable spatio-temporal learning framework for multi-client time-series prediction, personalized modeling, global/local pattern decomposition, and missing dependency recovery.

AMCT: Atom-Motif Contrastive Transformer for Molecular Property Prediction

ACM TIST 2026 first author

- Designed a Graph Transformer framework that jointly models atom-level and motif-level interactions for molecular representation learning.
- Introduced self-supervised atom-motif alignment and cross-molecule motif contrastive learning, with property-aware attention for interpretability.
- Completed systematic evaluation on 10 molecular property prediction benchmarks, forming a reusable pattern for graph Transformers, contrastive learning, and AI4Science workflows.

Open-World Visual Anomaly Detection System

Multimodal pretraining, real-time inference, project delivery

- Contributed to a visual anomaly detection system for public scenarios including malls, stations, streets, campuses, hospitals, and factories.
- Worked with CLIP / ActionCLIP-style multimodal pretraining, reverse auxiliary prompting, and class-imbalance semi-supervised learning for long-tailed anomaly events.
- The system achieved at least 25 FPS video stream processing, 97.9% F1 / detection success rate for typical anomalies, and around 0.25 s alert latency in project acceptance materials.

ConGCN / 2s-STGCN: Graph Neural Networks for Vision Applications

IEEE T-GRS 2023 first author; IEEE T-CYB 2022 student first author

- ConGCN integrated contrastive learning with graph convolution for hyperspectral image classification and became an ESI Highly Cited Paper.
- 2s-STGCN modeled first-order and second-order facial landmark dynamics with two-stream spatio-temporal GCNs for driver drowsiness detection, achieving 93.4% / 92.7% average accuracy on YawDD and NTHU-DDD.

PUBLICATIONS

1. Wentao Yu, Sheng Wan, Shuo Chen, Bo Han, and Chen Gong. Heterogeneity-Aware Knowledge Sharing for Graph Federated Learning. International Conference on Machine Learning (ICML), 2026. CCF-A Conference, First Author.
2. Wentao Yu, Shuo Chen, Yongxin Tong, Tianlong Gu, and Chen Gong. Modeling Inter-Intra Heterogeneity for Graph Federated Learning. AAAI Conference on Artificial Intelligence (AAAI), 2025. CCF-A Conference, First Author.
3. Wentao Yu, Sheng Wan, Guangyu Li, Jian Yang, and Chen Gong. Hyperspectral Image Classification with Contrastive Graph Convolutional Network. IEEE Transactions on Geoscience and Remote Sensing (IEEE T-GRS), 2023. CCF-B Journal, First Author, ESI Highly Cited Paper, CAS Q1.
4. Wentao Yu, Shuo Chen, Chen Gong, Bo Han, Gang Niu, and Masashi Sugiyama. Atom-Motif Contrastive Transformer for Molecular Property Prediction. ACM Transactions on Intelligent Systems and Technology (ACM TIST), 2026. CCF-C

Journal, First Author, JCR Q1.

5. Hang Zhou, Wentao Yu, Yang Wei, Guangyu Li, Sha Xu, and Chen Gong. Inter-Client Dependency Recovery with Hidden Global Components for Federated Traffic Prediction. AAAI Conference on Artificial Intelligence (AAAI), 2026. CCF-A Conference, Second Author.
6. Hang Zhou, Wentao Yu, Sheng Wan, Yongxin Tong, Tianlong Gu, and Chen Gong. Traffic Pattern Sharing for Federated Traffic Flow Prediction with Personalization. IEEE International Conference on Data Mining (ICDM), 2024. CCF-B Conference, Second Author, Best Student Paper Runner-Up Award.
7. Hang Zhou, Wentao Yu, Sheng Wan, Yongxin Tong, Tianlong Gu, and Chen Gong. FedTPS: Traffic Pattern Sharing for Personalized Federated Traffic Flow Prediction. Knowledge and Information Systems (KAIS), 2025. CCF-B Journal, Second Author, JCR Q2.

TECHNICAL SKILLS

- Programming: Python, C/C++, Linux, Git, shell-based experiment workflows.
- Machine Learning: PyTorch, PyTorch Geometric, NumPy, SciPy, scikit-learn, Matplotlib.
- Research Areas: Graph Neural Networks, Graph Transformers, Federated Learning, Personalized Federated Learning, Contrastive Learning, Self-Supervised Learning, Variational Inference, Spectral Graph Theory, Spatio-Temporal Modeling, Multimodal Learning, AI4Science.
- Experimentation: Multi-dataset benchmarking, ablation studies, hyperparameter management, reproducible code release, model evaluation, visualization, and technical writing.

AWARDS AND HONORS

- Best Student Paper Runner-Up Award, IEEE International Conference on Data Mining (ICDM), 2024.
- ESI Highly Cited Paper, IEEE Transactions on Geoscience and Remote Sensing, 2023.
- ESI Highly Cited Paper, IEEE Transactions on Cybernetics, 2022.
- Google Scholar 310+ citations.

OPEN SOURCE

- GitHub: <https://github.com/blgpb>
- Released research code and experimental configurations for FedIIH, ConGCN, FedGSP, and related graph/federated learning projects.
- Experience in preparing reproducible repositories with datasets, scripts, configuration files, and experimental documentation.

RESEARCH AND COLLABORATION STRENGTHS

- Strong ability to abstract real-world relational, temporal, and multimodal problems into graph-based or structured machine learning formulations.
- Experience in cross-institution collaboration with researchers from Nanjing University of Science and Technology, Hong Kong Baptist University, and RIKEN AIP.
- Solid publication record across top-tier AI/ML conferences and high-impact journals, with first-author work in ICML, AAAI, ACM TIST, and IEEE T-GRS.